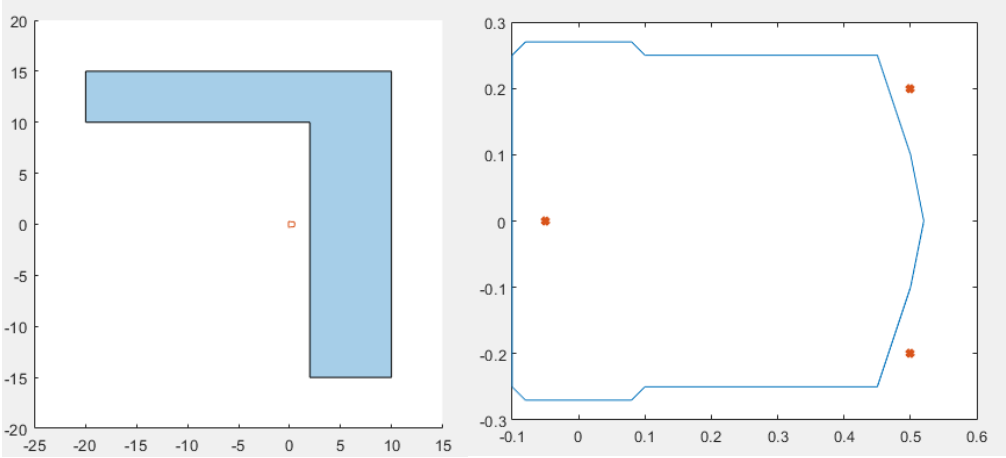
Homework Assignment for the Course "Current Research Areas II "

You were already introduced to the iMOW robotic lawnmower and some of its functions. Now, a Matlab/Simulink model of the abstracted robot model (mobile robot with differential drive) should be created, and the so-called "Random-Walk" should be implemented.

The implementation should be done in groups, and the task will be graded based on a final presentation where you explain the algorithm you developed and present the implementation and simulation results.

Description:

- Mobile robot with differential drive, as introduced beforehand
- Mowing area as described
- Coil arrangement according to the sketch
  - 

- Input variables are:
  - Translational speed with $\mathrm{v_{max}} = \pm\, 0.5\,\frac{\mathrm{m}}{\mathrm{s}}$
  - Rotation speed with $\omega_{max} = \pm\, 1\,\frac{rad}{s}$
- The control unit on the mower operates at a sampling rate of $\mathrm{T_s} = 10\mathrm{ms}$
- Simulation time = 6000s

Task:

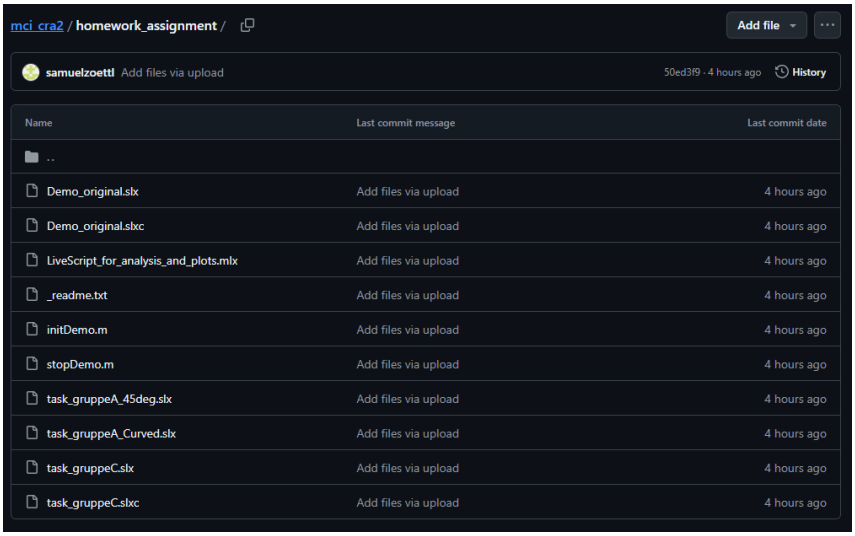
- Implement the mathematical model as a nonlinear state-space model.
- Simulate the model in Simulink.
- Implement a typical garden geometry (hexagon with right angles = L-shape).
- Implement the coil signals of the robotic mower.
- The output signal should indicate "Inside" within the garden and "Outside" outside the field.
- Implement a state-flow chart with at least the following functions:
  - "Forward"
  - "Backward"
  - "Right Turn"
  - "Left Turn"

    - "Stop"
- Ensure that the mower can only operate within the garden aka Working Area (at least one coil "Inside").
    - No sensor coil should stay outside of the working area for more than 1 seconds
    - One Coil has always to be inside
    - Formulate requirements for the different involved “software modules” and also note, how you want to test against them
    - Use the state-flow chart and coil signals for the implementation
- Develop an algorithm to ensure the entire garden is mowed.
    - Only use the coil signals as input signals.
    - How do you implement “mowing” (what area is mowed, when the mower travels through the Working Area)
    - Think of ways to measure the progress and efficiency of your mowing logic
    - When do you abort your mowing session?
    - Formulate requirements for the different involved “software modules” and also note, how you want to test against them
    - Implement tests to ensure your most critical requirements are fulfilled
- Extend the Simulink model with your algorithm.
- Simulate the extended model
- Present proof for the efficiency (eg create a plot; show data) of the algorithm.
    - You could plot the % of mowed area against time and compare algorithms

HOMEWORK ASSIGNMENT:

- Form of submission:
  - Working folder (based on assignment files) with models and live script
  - Summary of your results. Presentations are preferred

- Find the files for the assignment here: (Consider the tasks above as implemented)
  mci_cra2/homework_assignment at main · samuelzoettl/mci_cra2
  we base our homework assignment on the results from the group sessions.



  - Short explanation:
    - initDemo and stopDemo are called from within the Simulink model
    - we use the LIVE Script "LiveScript_for_analysis_and_plots.mlx" to analyze the coverage and plot some data
    - the models in the folder are: Demo_original, task_gruppeA_45deg, task_gruppeA_Curved, task_gruppeC

- homework assignment Task 1:
  - Analyzing the results of the group work, we can conclude that the mower does not mow the areas close to the borders of the area. This results in unattractive edges, a problem also very common in real live lawn mower robots. The reason is quite simple: The cutting device is located more towards the center of the robot, whereas the sensor coils for the perimeter wire detection are located in the front of the machine.
  - Use a chosen mower logic (from the included models) and adapt the behavior on the edges of the working area. ("drive a bit farther over the wire before turning")
    - Try to stay within the "rules" (No sensor coil should stay outside of the working area for more than 1 seconds; One Coil has always to be inside)
    - **Compare the base models area coverage to your extended model**
    - **Discuss briefly if the 1 second rule from above limits your mowing strategy and why. What happens if you ignore the rule?**

- homework assignment Task 2:
    - same challenge as above, different approach!
    - Add a separate maneuver to a chosen mower logic (from the included models). Let the robot follow the perimeter wire and mow the edge (use the sensor coils to drive along the wire) before you start with the established mowing logic. You will need a condition to end this edge mowing operation, that works independently from the shape of the garden.
        - **Compare the base models area coverage to your extended model**
        - **Discuss briefly how you determine the edge mowing maneuver to be finished**